

Principal Component Analysis

Ian Castleton

July 9, 2019

1 Introduction

PCA is a dimensionality reduction technique which finds "important directions" (basis vectors which capture most variation of the data) inherent in a dataset. It projects data points into these basis vectors to obtain a low dimensional representation of the data points. In this document we consider two examples which illustrate the features of PCA: i) A simple 2D (x,y) dataset, followed by ii) A more complex bond yield example.

2 Simple 2D dataset

The simple 2D data set example follows the article: <https://stats.stackexchange.com/questions/229092/how-to-reverse-pca-and-reconstruct-original-variables-from-several-principal-com>. For this example the raw input data is constructed from random noise superimposed on the line $y = x$. These (x, y) values form a matrix called X_{raw} . The data is plotted in figure 1.

2.1 Standarising the input

The PCA approach works best when the input data has no trend (i.e. an average level of zero) and when all the data dimensions have similar range. This process is known as standarising the input data. In this simple example it will be sufficient to centre the data by subtracting the average. We shall refer to this standardised data by the matrix X . The standardised data is plotted in figure 2.

2.2 Performing the PCA calculation

The MLIBQ PCA function is called: `meUtilityPCA()`. Internally the PCA function computes the covariance (or optionally the correlation) matrix of the input data, then computes the eigenvectors of that matrix. The input parameters are:

- **Key:** Any unique name to identify this calculation and its results
- **Data:** The standardised block of input data
- **UseCorrelationMatrix:** If set to true, calculate eigenvectors of the correlation matrix. Otherwise calculate the eigenvectors of the covariance matrix. The practical difference between the two is that the correlation matrix is

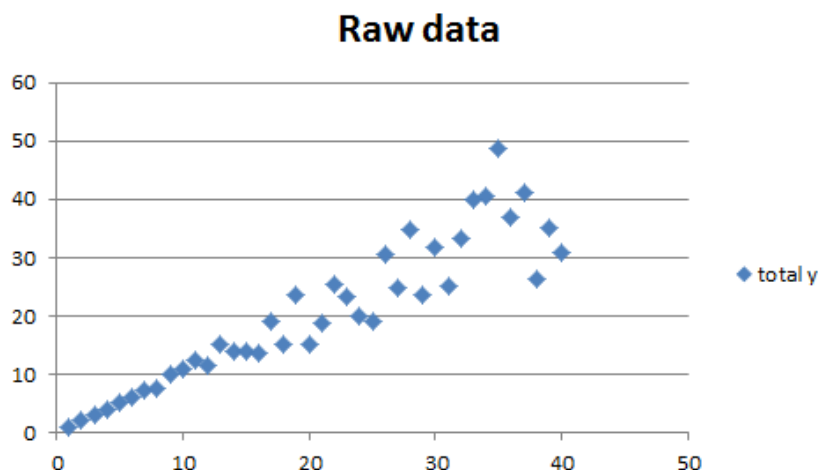


Figure 1: Raw data

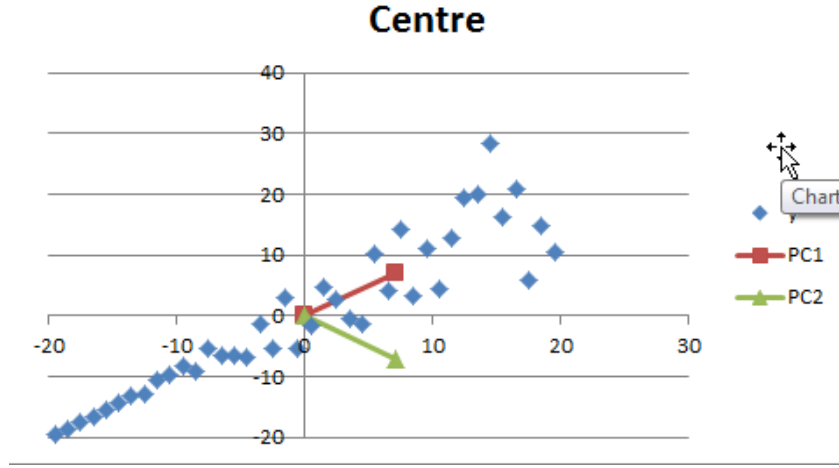


Figure 2: Standardised data which is centred on the origin

normalised such that the correlations lie within the range $[-1, +1]$. If the input data has a widely different range for each dimension, better results will be obtained using the correlation matrix.

- **NFactors:** Specifies the number of PCA factors (eigenvectors) to calculate for the input data. This must be equal to or less than the number of dimensions in the input data. In this example we set **NFactors** equal to 2.

Below is the output from running the PCA analysis on the input data. Since **nFactors=2**, we obtain 2 eigenvectors:

	PC1	PC2
Series1	0.71	0.71
Series2	0.71	-0.71
POV (Proportion of Variance)	96%	4%
EigenValues	192%	8%

The output consists of the following sections:

- The column labels PC1 and PC2 refer to the computed eigenvectors. One of the properties of PCA is that these eigenvectors are orthonormal. i.e. each vector has unit length, and the eigenvectors are orthogonal to each other.
- The labels Series1 and Series2 correspond to the two dimensions (x and y) of the input data.
- PC1 is an eigenvector with an x-component of 0.71 and a y-component of 0.71. i.e. this is a unit vector at 45 degrees to the x axis. Similarly PC2 is an eigenvector with x-component of 0.71 and a y-component of -0.71. i.e. this is a unit vector at -45 degrees, at right-angles to the first vector. This satisfies the orthogonality condition.
- POV means "Proportion of variance". It describes how much of the input data variance is explained by each eigenvector. Another key feature of the PCA is that the eigenvectors are sorted in order of decreasing POV, so that PC1 always accounts for the majority of the data variance. In this example we see that eigenvector PC1 explains 96% of the input data variance.
- The EigenValue row numbers are similar to the POV numbers without the normalization to 100%.

Figure 2 also plots the PC1 and PC2 eigenvectors alongside the standardized input data. We see that PC1 (red line) follows the general trend of the data ($y = x$) while PC2 is at right angles, measuring how much each data point lies to the side of the general trend line. The PCA algorithm ensures that these eigenvectors are aligned with the principal axes of the data set. Mathematically we shall refer to the eigenvectors by the matrix V .

2.3 PCA Scores

The matrix of "PCA scores" Z is defined as follows:

$$Z = XV \quad (1)$$

i.e. the standardized input data is multiplied by the eigenvector matrix. Each score is a projection of the data points onto each eigenvector / principal axis. The score can be considered as the distance of each data point along each principal axis. Figures 3 and 4 below show the scores obtained for PC1 and PC2. Note that these points are not necessarily in order. Through this transformation of X into Z , the eigenvector columns of V are sometimes known as "weight vectors" or "loading vectors", since the columns of this matrix specify how much of each X point is used to construct the corresponding Z point.

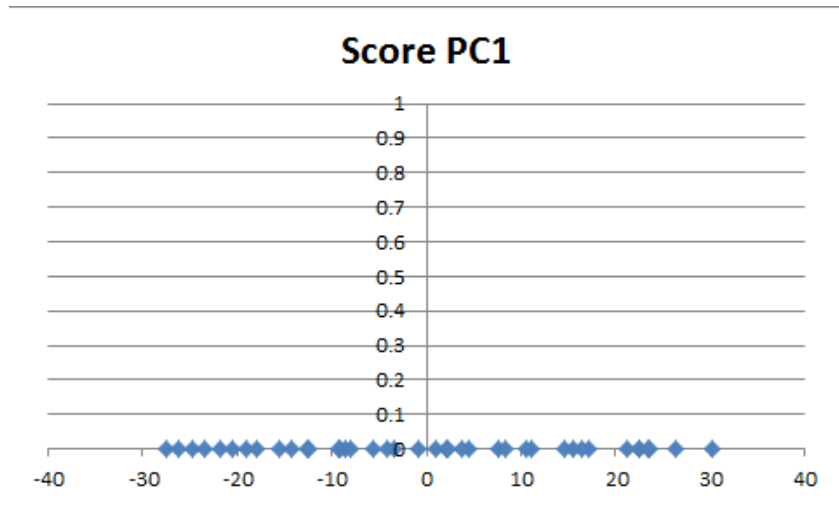


Figure 3: Calculated score using PC1

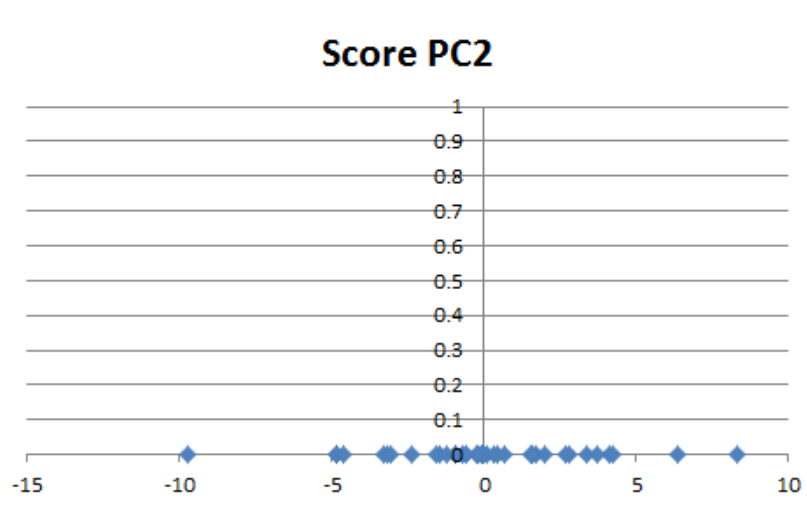


Figure 4: Calculated score using PC2

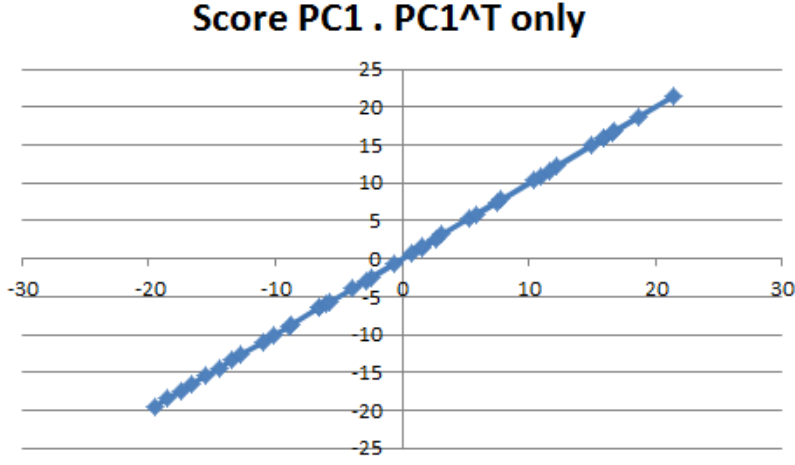


Figure 5: Data reconstruction from PC1 score

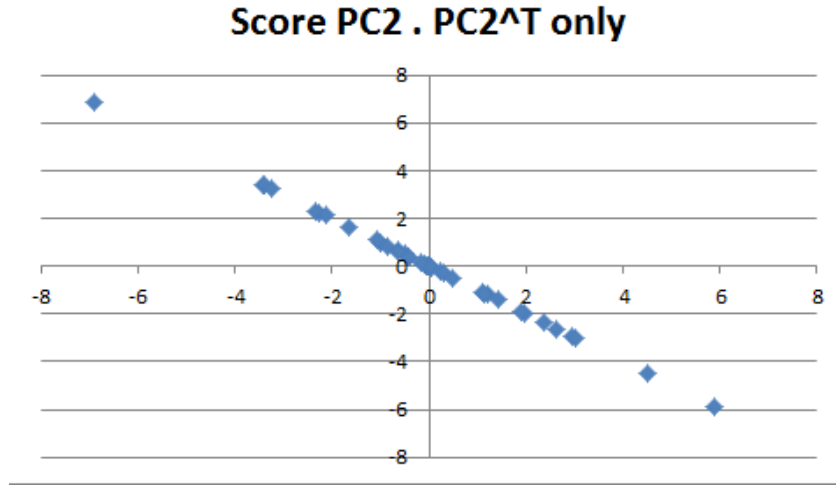


Figure 6: Data reconstruction from PC2 score

2.4 PCA Reconstruction

In order to reconstruct the original two variables from each principal component, The scores are mapped back into the original (x,y) dimensions by multiplying by W^T , a reduced matrix containing only the eigenvalues of interest. This is known as the reconstruction R . Note that V is adjusted to include either PC1 or PC2, not both:

$$R = ZV^T = XW^T \quad (2)$$

Using the full eigenvector matrix would actually yield the original data X , since VV^T is the identity matrix. Figures 5 and 6 show the reconstructions obtained after multiplying by a transposed matrix containing only PC1 or PC2.

2.5 Reduced dimensionality

Since the majority of the variance (96%) is explained using PC1 only, we reconstruct each data point using PC1 and discard PC2. The gradient of PC1 is given by $PC1_y/PC1_x$. Hence the reconstructed y values are:

$$\hat{y} = x (PC1_y/PC1_x) \quad (3)$$

Figure 7 plots these y values, while figure 8 plots the residual $(y - \hat{y})$ between the original y values and these new reconstructed values.

3 US Treasury Bond yields

In this example we analyse historical US Treasury bond yields. We follow closely the article described here: <https://insightr.wordpress.com/2017/04/14/american-bond-yields-and-principal-component-analysis>. The input

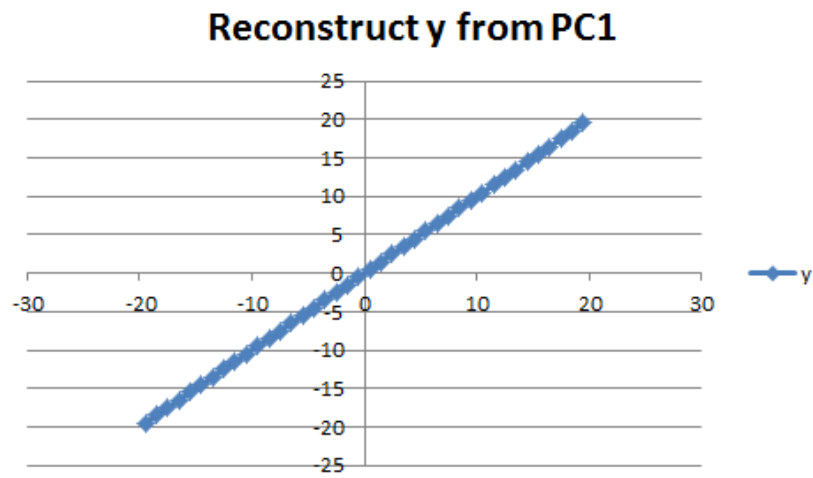


Figure 7: Reconstruct y data using PC1 only

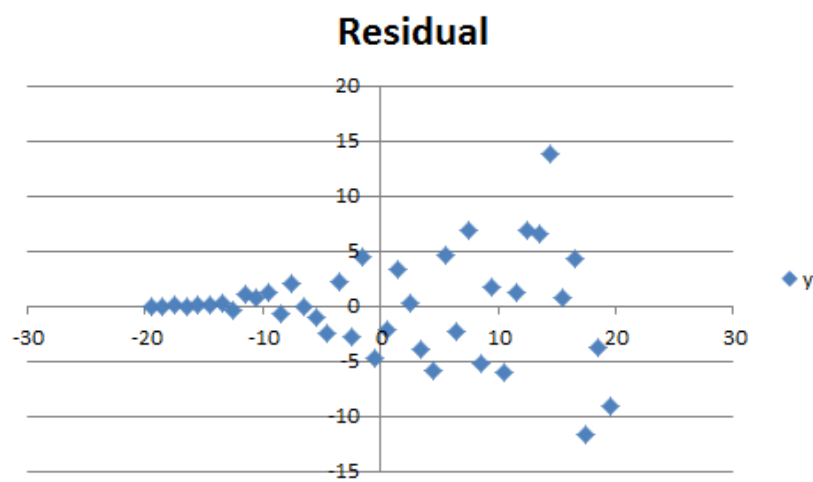


Figure 8: Residual noise

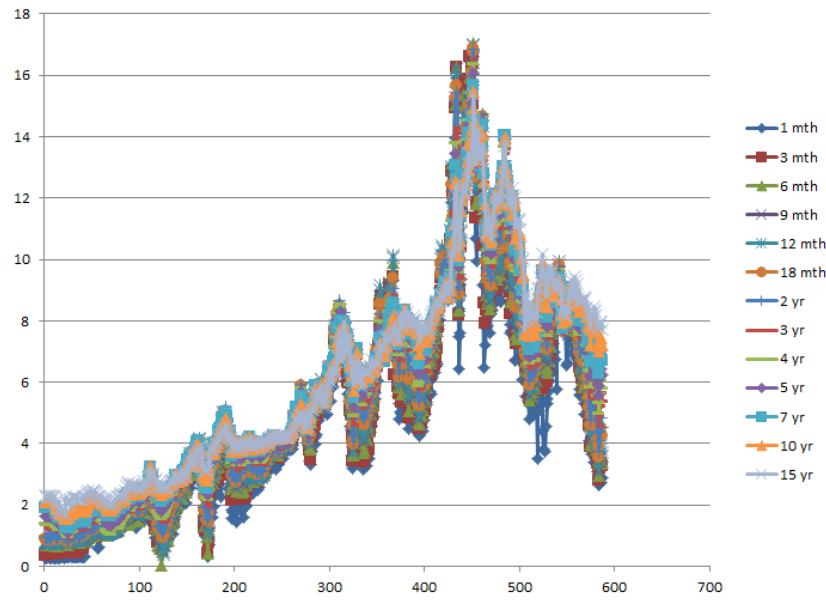


Figure 9: Historical bond yields

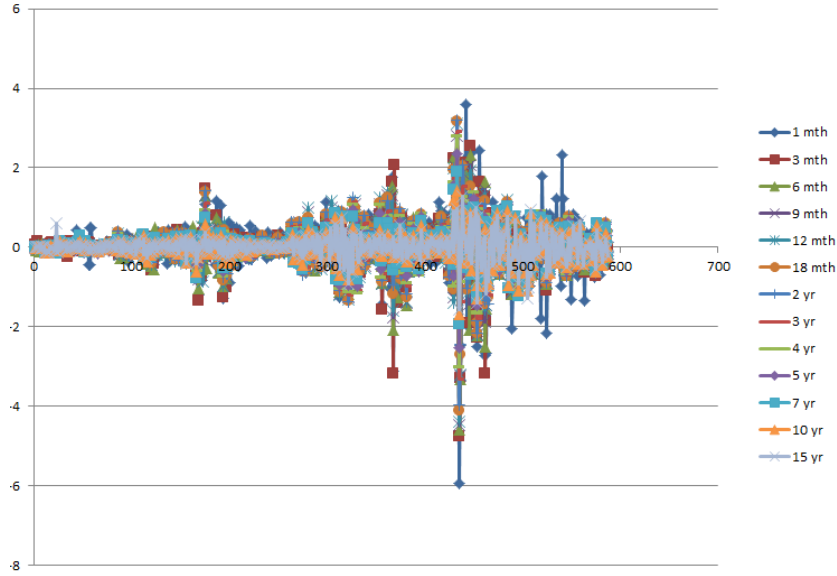


Figure 10: Monthly Yield Returns

data consists of annualised bond yields for bonds with maturities in the range 1 month to 15 years. The data set contains a monthly yield point for each bond maturity, resulting in a time-series of bond yields. The raw yield data is plotted in figure 9. Although yields for different bond maturities have a strong correlation structure, they are not good for modelling because they are not stationary. As with the previous example, the PCA approach requires data with no long term trend. Instead, it turns out that using the yield *returns* is appropriate, since the returns tend to hover around zero with no long term drift. We calculate the i th monthly return for bond k as follows:

$$R_{i,k} = \text{Yield}_{i,k} - \text{Yield}_{i-1,k} \quad (4)$$

Figure 10 plots the monthly yield returns. Applying the function `meUtilityPCA()` on this data produces the eigenvector results shown in figure 11. Note that there are 13 eigenvectors, since the input time-series data contains 13 dimensions (13 bonds). Note furthermore that the first eigenvector PC1 accounts for 82% of the variance, and the first three eigenvectors in total account for 96% of the variance. Figure 12 plots the first three eigenvectors. Three observations can be made from this eigenvector plot:

1. PC1 is an approximately horizontal line. This means that the dominant feature of the input data is for all bond returns to move in the same direction together.
2. PC2 is a monotonically decreasing straight line: The short dated bonds all move in one direction while the longer dated bonds tend to move in an opposite direction. This implies that the bonds can be partitioned into two clusters

Output	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11	PC12	PC13
Series1	0.20	0.48	0.68	0.45	0.25	0.03	0.08	0.01	0.00	0.00	0.00	0.00	0.00
Series2	0.25	0.42	0.10	-0.30	-0.59	-0.31	-0.38	-0.29	0.00	0.00	0.00	0.00	0.00
Series3	0.28	0.30	-0.08	-0.31	-0.21	0.10	0.53	0.57	-0.01	0.00	0.06	0.28	0.01
Series4	0.29	0.21	-0.19	-0.18	0.14	0.30	0.15	-0.12	0.01	0.01	-0.18	-0.79	-0.02
Series5	0.29	0.16	-0.24	-0.12	0.31	0.40	-0.05	-0.48	-0.03	-0.05	0.39	0.41	0.01
Series6	0.30	0.06	-0.21	0.01	0.29	-0.06	-0.29	0.11	0.05	0.12	-0.75	0.31	0.01
Series7	0.30	0.01	-0.19	0.08	0.27	-0.31	-0.41	0.44	0.00	0.07	0.46	-0.17	-0.29
Series8	0.30	-0.11	-0.17	0.22	0.02	-0.32	0.10	-0.01	-0.07	-0.39	0.07	-0.06	0.74
Series9	0.29	-0.17	-0.15	0.30	-0.12	-0.31	0.41	-0.29	-0.15	-0.22	-0.10	0.04	-0.58
Series10	0.29	-0.22	-0.04	0.27	-0.21	0.01	0.14	-0.10	0.16	0.81	0.09	-0.01	0.18
Series11	0.28	-0.28	0.11	0.21	-0.33	0.45	-0.23	0.15	0.51	-0.35	-0.01	0.01	-0.09
Series12	0.27	-0.35	0.28	-0.11	-0.11	0.26	-0.17	0.11	-0.77	0.02	-0.03	0.00	0.02
Series13	0.24	-0.38	0.45	-0.55	0.30	-0.27	0.14	-0.12	0.32	-0.01	0.01	0.00	-0.01
POV	82%	11%	3%	2%	1%	1%	0%	0%	0%	0%	0%	0%	0%
Eigen Values	10.6067	1.44852	0.403437	0.259172	0.135755	0.067287	0.043053	0.035602	0.000191	0.000103	7.44E-05	5.31E-05	9.35E-05

Figure 11: Bond Yield-return Eigenvectors

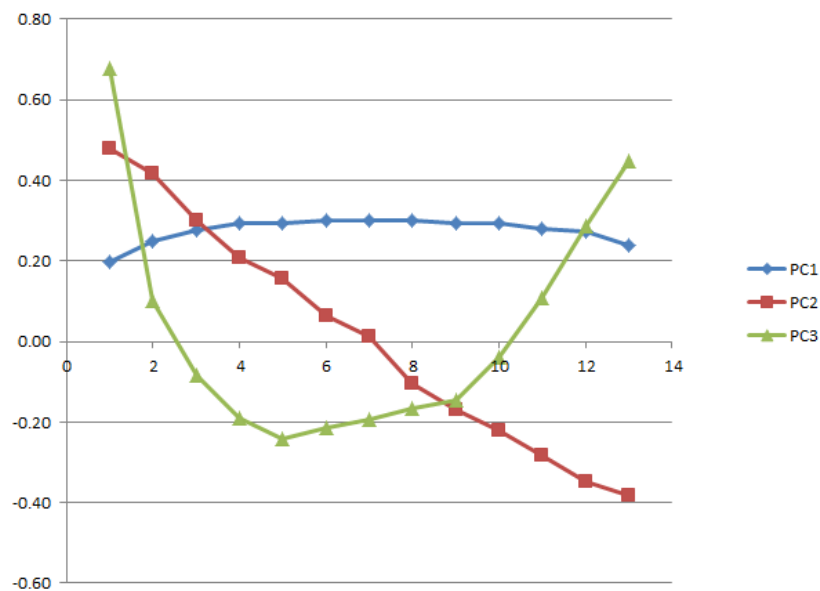


Figure 12: Plotting the first three eigenvectors: Parallel shift, slope, convexity



Figure 13: Simulation of bond yields 5 months into the future. The simulation is performed for a 2 year and 10 year maturity US Treasury.

based on maturity date. The intuition behind this observation is that short data bonds will be more affected by changes in central bank base rate.

3. PC3 shows a convexity behaviour: Very long and short dated bonds move together, while bonds with mid-dated maturities move in an opposite direction.

3.1 Using PCA for Prediction

As in the previous example, to recover the yield returns from the principal component scores we multiply by a reduced eigenvector matrix containing only the eigenvectors of interest.

$$R = ZW^T \quad (5)$$

Once we have yield returns, the original yields can be reconstructed from

$$\text{Yield}_{i,k} = R_{i,k} + \text{Yield}_{i-1,k} \quad (6)$$

This then gives a way to predict the future spread of bond yields. The impact on yield returns for the first, second and third principal components are given by

$$R_{t+1,*} = R_{t,*} \pm \sqrt{\lambda_1} W_{*,1} \quad (7)$$

$$R_{t+1,*} = R_{t,*} \pm \sqrt{\lambda_2} W_{*,2} \quad (8)$$

$$R_{t+1,*} = R_{t,*} \pm \sqrt{\lambda_3} W_{*,3} \quad (9)$$

where $W_{*,k}$ represents the k -column of the eigenvalue matrix and $\sqrt{\lambda_k}$ represents the standard deviation of the k -principal component. The plot in figure 13 shows the possible yield paths for a 2 year and 10 year maturity swap, over the next 5 months. The plots were produced by performing a monte-carlo simulation.

4 Summary

1. PCA is used to reduce dimensionality and overcome feature redundancy in a data set.
2. These features are low dimensional in nature i.e. the dataset can be explained by very few eigenvectors.
3. These features a.k.a component scores are a result of normalized linear combination of original predictor variables.
4. These components aim to capture as much information as possible with high explained variance.
5. The first component has the highest variance followed by second, third and so on.

6. The components (eigenvectors) are orthogonal i.e. uncorrelated.
7. Normalizing data becomes extremely important when the predictors are measured in different units.
8. PCA works best on data set having 3 or higher dimensions, which would otherwise have too many dimensions to visualise.